C# Programming Fundamentals

Assignment 1 — Sample Answer

Objective

This sample console application demonstrates variables, data types, conditional statements, loops, arrays, methods, input validation, and exception handling. It calculates a student's total marks, average, and grade.

C# Code

```
using System;

namespace StudentMarksCalculator
{
    internal class Program
    {
        static void Main()
        {
            try
            {
                Console.Write("Enter student name: ");
                string name = Console.ReadLine() ?? "";

                if (string.IsNullOrWhiteSpace(name))
                {
                    Console.WriteLine("Student name cannot be empty.");
                    return;
                }

                const int SubjectCount = 3;
                int[] marks = new int[SubjectCount];
                string[] subjects = { "C#", "Database", "Computer Networks" };

                for (int i = 0; i < SubjectCount; i++)
                {
                    Console.Write($"Enter marks for {subjects[i]} (0-100): ");
                    string input = Console.ReadLine() ?? "";

                    if (!int.TryParse(input, out marks[i]))
                    {
                        Console.WriteLine("Invalid input. Please enter a whole number.");
                        return;
                    }

                    if (marks[i] < 0 || marks[i] > 100)
                    {
                        Console.WriteLine("Marks must be between 0 and 100.");
                        return;
                    }
                }

                int total = CalculateTotal(marks);
                double average = (double)total / marks.Length;
                string grade = CalculateGrade(average);

                Console.WriteLine();
                Console.WriteLine($"Student: {name}");
                Console.WriteLine($"Total: {total}/{SubjectCount * 100}");
                Console.WriteLine($"Average: {average:F2}");
                Console.WriteLine($"Grade: {grade}");
            }
            catch (Exception ex)
            {
                Console.WriteLine($"An unexpected error occurred: {ex.Message}");
            }
        }

        static int CalculateTotal(int[] marks)
```

```
        {
            int total = 0;

            foreach (int mark in marks)
            {
                total += mark;
            }

            return total;
        }

        static string CalculateGrade(double average)
        {
            if (average >= 90)
                return "A";
            else if (average >= 80)
                return "B";
            else if (average >= 70)
                return "C";
            else if (average >= 50)
                return "D";
            else
                return "F";
        }
    }
}
```

Concepts Demonstrated

• **Variables and data types:** string, int, double, and const.

• **Conditional statements:** if / else if / else for input checks and grade calculation.

• **Loops:** for loop to collect marks and foreach loop to calculate the total.

• **Arrays:** stores marks and subject names.

• **Methods:** CalculateTotal and CalculateGrade keep logic reusable.

• **Input validation:** int.TryParse and checks that marks are between 0 and 100.

• **Exception handling:** try / catch handles unexpected runtime errors.

Sample Run

```
Enter student name: Alex
Enter marks for C# (0-100): 92
Enter marks for Database (0-100): 84
Enter marks for Computer Networks (0-100): 88

Student: Alex
Total: 264/300
Average: 88.00
Grade: B
```

Note: This is sample content for testing the assignment PDF upload workflow. You can upload this file as a student's submission in your application.